

Qatar-1b

R-2600

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110926 · created 2026-08-28T00:29:46+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455829.39667 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.66 +/- 2.16 [%]
Orbital Inclination (inc)	83.66 +/- 2.89
Airmass coefficient 1 (a1)	0.4701 +/- 0.0043
Airmass coefficient 2 (a2)	0.51 +/- 0.59
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#3 - [342, 468]
Optimal Aperture	7.99
Optimal Annulus	18.65
Transit Duration (day)	0.0222 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.056 vs 0.1463 (deviation 0.8 σ)
Duration fit vs published	0.0222 d vs 0.0692 d (deviation 7.12 σ)
Mid-time O-C	-2044.1 min
Depth SNR	3.4
Residual scatter	0.92 %

Light curve

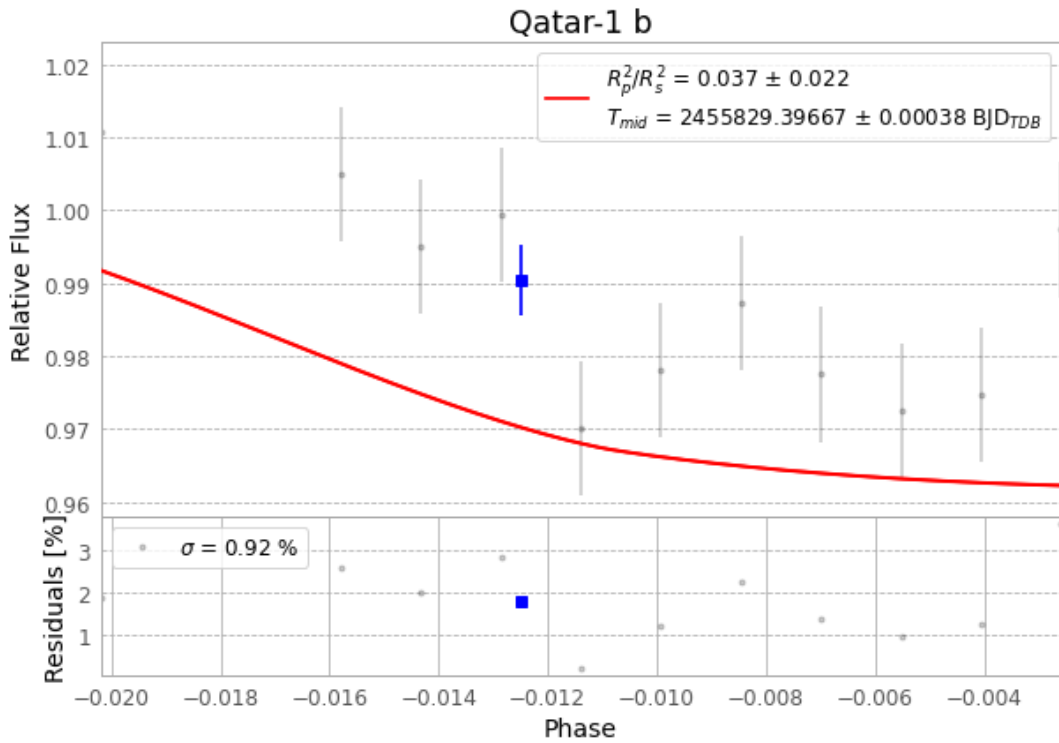


Figure 1 — FinalLightCurve_Qatar-1 b_2011-09-25.png (SHA-256 5880b45bc6abcf78...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [426, 296], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee0467c19e58d677...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

49 FITS frames (32 MB), Qatar-1110926065112.FITS → Qatar-1110926092713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110926.log (9eced07ecff8...), FinalLightCurve_Qatar-1 b_2011-09-25.png (5880b45bc6ab...), FinalLightCurve_Qatar-1 b_2011-09-25.csv (fc28e7565c4b...)

Compute resources

Wall time 379.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:29Z.